خدمتكم طريق خضناه لرضى الله

2021

أسئلة سنوات فاينال

مختبر كيمياء عامة





سنافر البوليتكنك | 0790640139

بسم الله الرحمن الرحيم

نقدم لكم نحن أسرة فريق

(لجنة السنافر)

مجموعة أسئلة اختبارات إلكترونية تم تجميعها خلال الفصول الماضية
سائلين المولى أن يوفقنا وإياكم لكل خير

تنويه

يوجد بعض الأسئلة عليها إجابات قد تحتمل الصواب وقد تحتمل الخطأ

فإن أصبنا فما هو إلا توفيق من الله

وإن أخطأنا فمن أنفسنا

#خدمتكم_طريق_خضناه_لرضى_الله

#الإتجاه_الاسلامي

#بسواعدنا_نبنيها

#لجنة_السنافر

#هي_لله

A pipette is used to dispense small precise quantity of liquid.



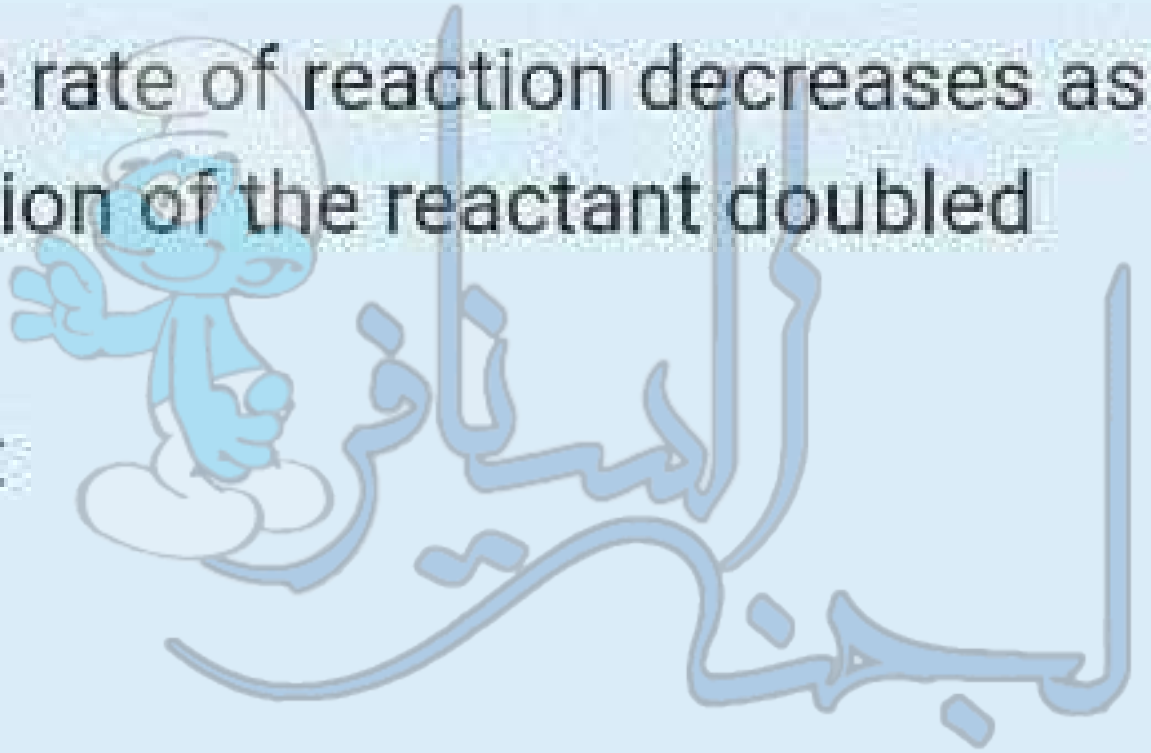
Select one:

- True
- False

Usually the rate of reaction decreases as the concentration of the reactant doubled

Select one:

- True
- False



.All these salts are hydrated

اختر أحد الخيارات

صح

خطأ

$KAl(SO_4)_2 \cdot 12H_2O$

$KCr(SO_4)_2 \cdot 12H\ 2O$

$Na_2B_4O_7 \cdot 10H_2O$

If some glass impurities contaminate the liquid the boiling point of liquid does not change.

اخترأحد الخيارات

- صح
- خطأ

When all of the liquid was vaporized when the flask was removed from the water bath during molecular weight of volatile liquid experiment then the calculated molar mass to be more.

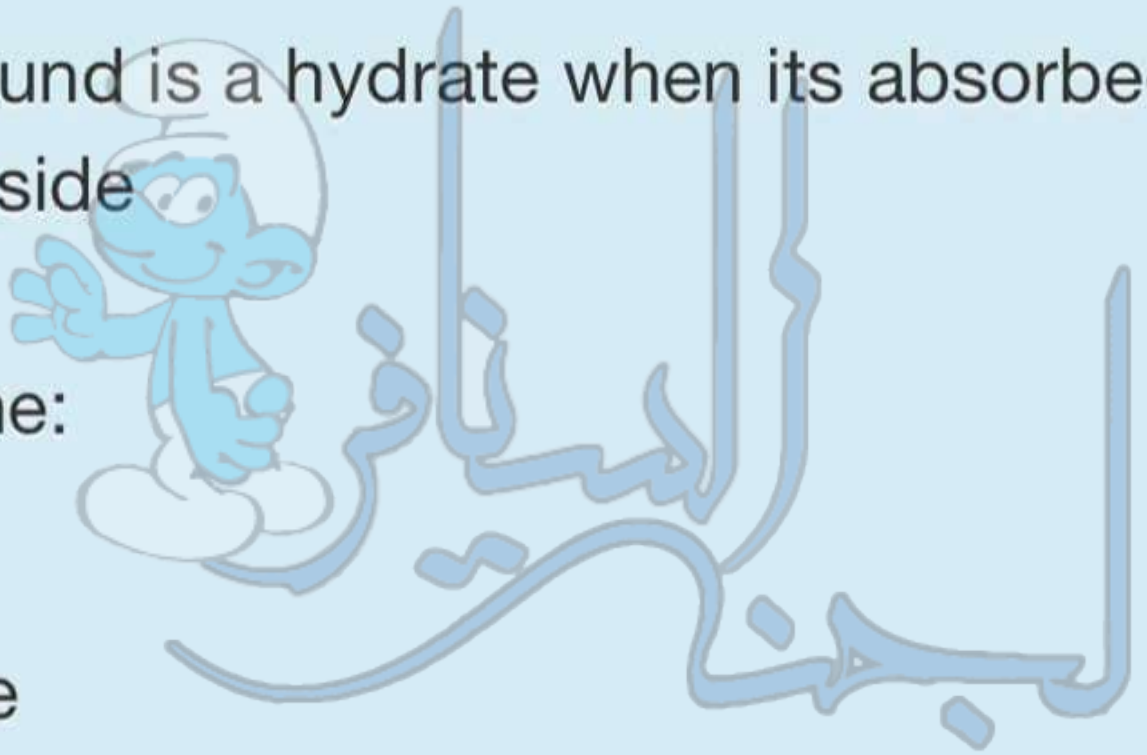
اخترأحد الخيارات

- صح
- خطأ

A compound is a hydrate when its absorbed water from out side

Select one:

- True
- False



When a substance is dissolved in another substance, it is said to be soluble.

Select one:

- True
- False

All sodium and potassium salts are water soluble.

اختر أحد الخيارات

- صح
- خطأ



Using water bath while heating the sample of volatile liquid is one of the true step during molecular mass of volatile liquid experiment procedure.

اخترأحد الخيارات

- صح
- خطأ

.Water will show two layers with Hexane

اخترأحد الخيارات

- صح
- خطأ



An Alcohol has 8 carbons will make one layer solution with Hexane.

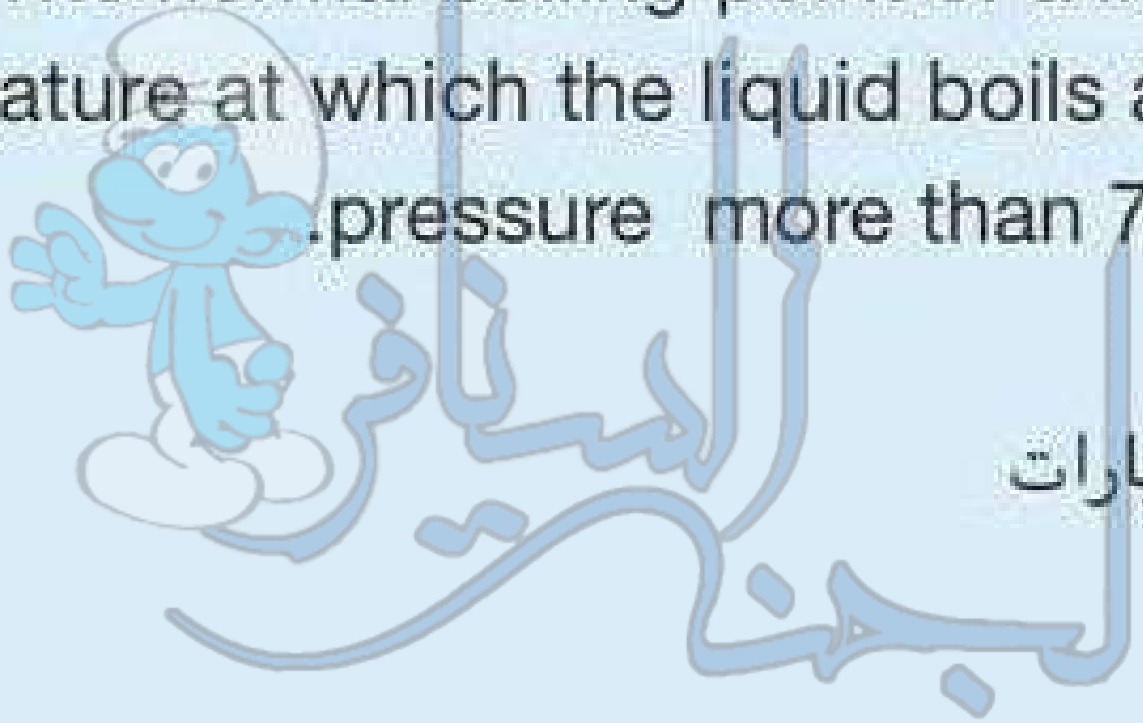
Select one:

- True
- False

The normal boiling point of a liquid is the temperature at which the liquid boils at external pressure more than 760 mmHg.

اخترأحد الخيارات

- صح
- خطأ



Hypochlorite ion OCl^- is the bleaching agent in bleach.

Select one:

- True
- False



Methane (CH_4) contain weak bond so its boiling point will be lower than Ethanol (CH_3OH).

Select one:

- True
- False



Methane (CH_4) contain weak bond so its boiling point will be lower than Ethanol (CH_3OH).

Select one:

- True
- False

The difference between a hydrous and an anhydrous salt is that the hydrous is an anhydrous with water removed from.

اخترأحد الخيارات

- صح
- خطأ

The normal boiling point is specified and tabulated when the vapor pressure of liquid is 1.0 atm.



Select one:

- True
- False

For the reaction which m^{th} order with respect to A and n^{th} order with respect to B; its overall order is m + n



اخترأحد الخيارات

- صح
- خطأ

For the reaction which m^{th} order with respect to A and n^{th} order with respect to B; its overall order is m + n



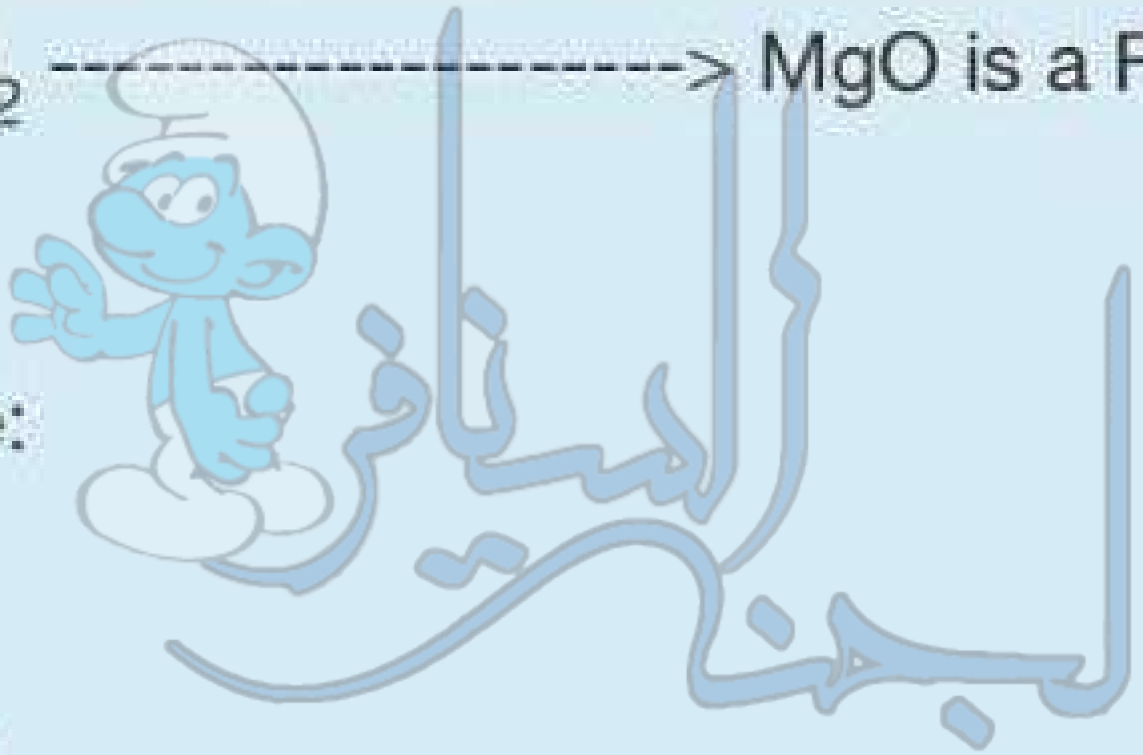
اخترأحد الخيارات

- صح
- خطأ

$Mg + 1/2O_2$ ----------> MgO is a Formation reaction

Select one:

- True
- False



In bleach titration experiment the color at the end point is yellow

اخترأحد الخيارات

- صح
- خطأ

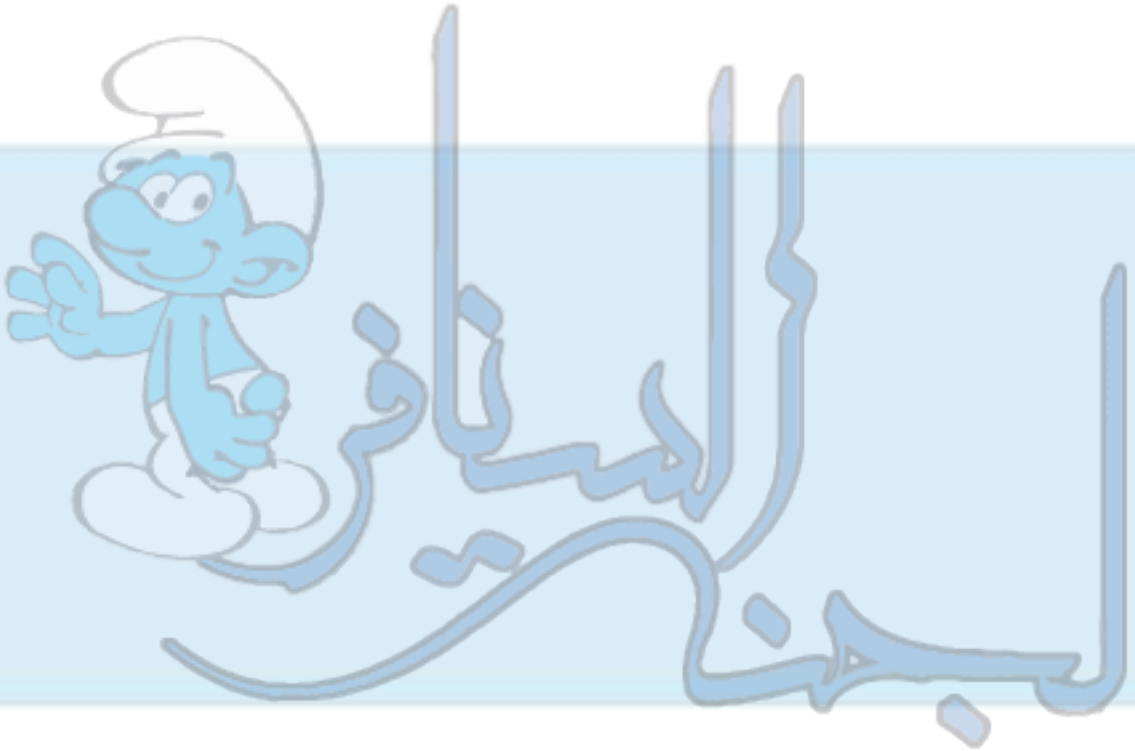


All these salts are alums.

Select one:

- True
- False

$KAl(SO_4)_2 \cdot 12H_2O$
$KCr(SO_4)_2 \cdot 12H\ 2O$
$Na_2B_4O_7 \cdot 10H_2O$



We can use Dumas method for any liquid whose boiling point is less than 92 c° to determine its density



اخترأحد الخيارات

- صح
- خطأ

In water of crystallization experiment, You need to add water to salt so you calculate the value of (X) correctly

اخترأحد الخيارات

- صح
- خطأ

Desiccators used to protect the chemicals compounds which are hygroscopic or which react with water from humidity.



اختر أحد الخيارات

- صح
- خطأ

In water of crystallization experiment, You need to add water to salt so you calculate the value of (X) correctly

اخترأحد الخيارات

- صح
- خطأ

Endothermic reaction has a positive sign of enthalpy (ΔH +ve)



اختر أحد الخيارات

- صح
- خطأ

Diethyl ether has a dipole-dipole force so that it has a higher boiling point than water H_2O.



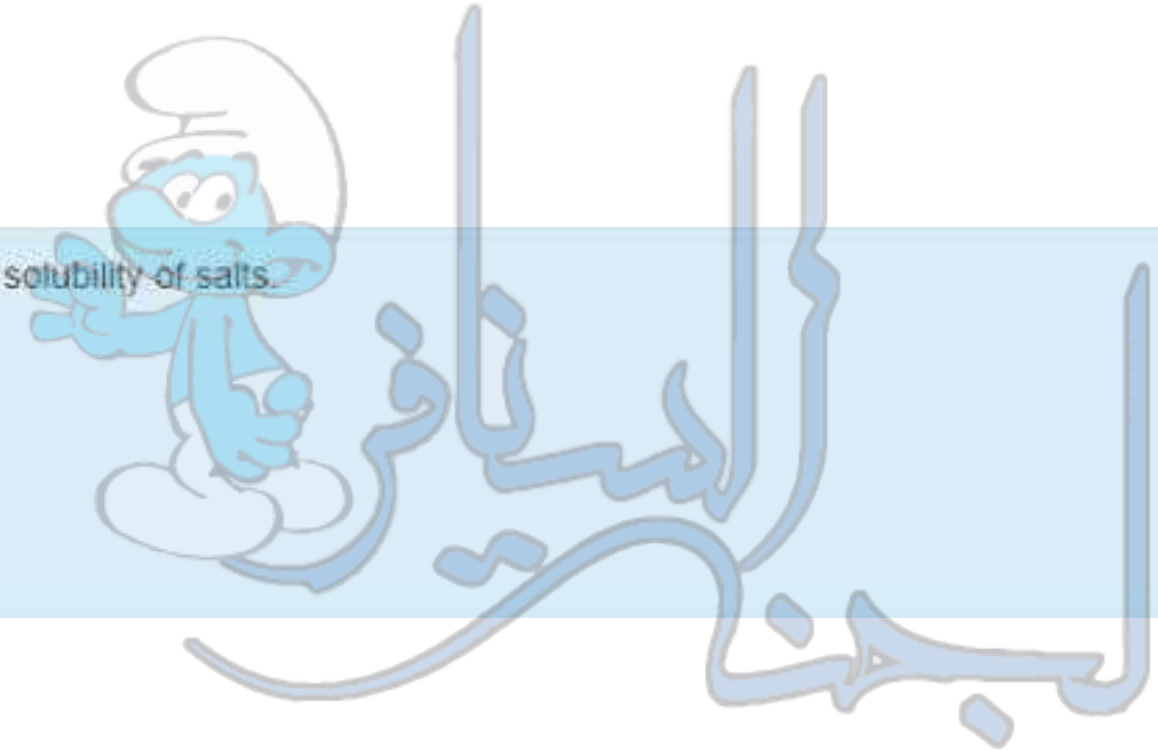
Select one:

- True
- False

Cooling will decreasing the solubility of salts.

Select one:

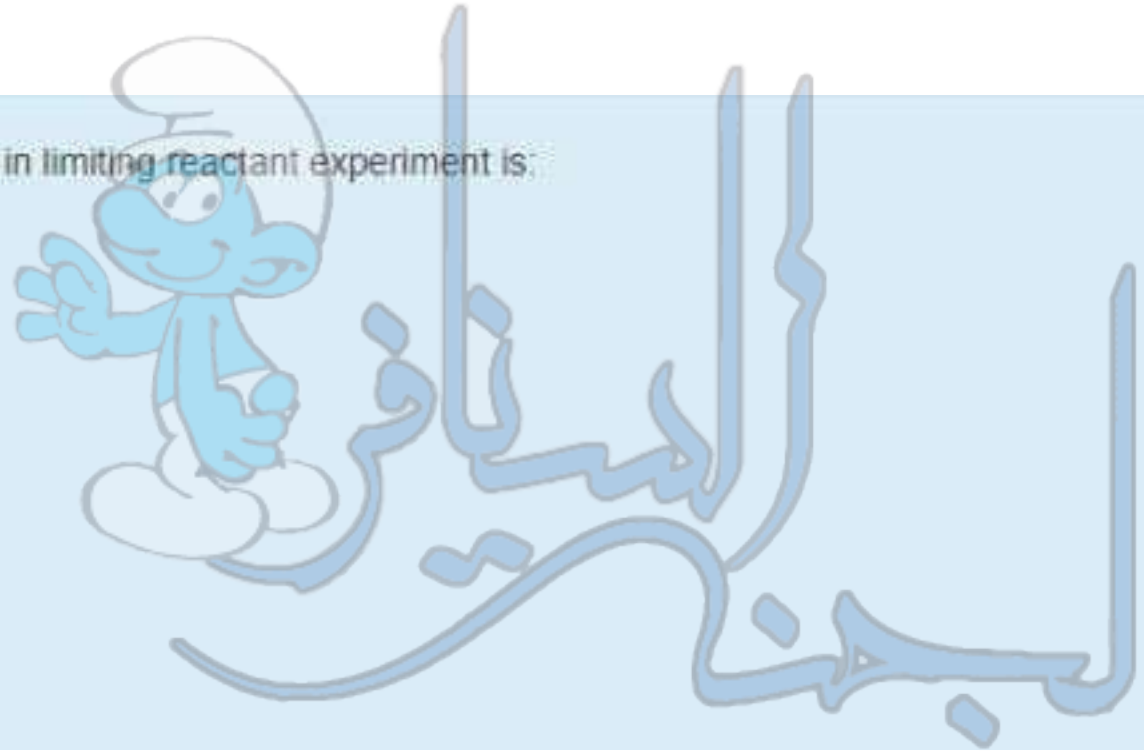
- True
- False



The color of the precipitate in limiting reactant experiment is:

Select one:

- a. Blue
- b. White
- c. Brown
- d. Pink



When the flask was not dried before the final weighing with the condensed vapor inside then the calculated molar mass would be lower.

Select one:

- True
- False

Taking the temperature of boiling water after the flask cooling down will affected the calculated molar mass of volatile liquid to be higher.

Select one:

- True
- False

A suitable glassware used to determine the volume of the gas in volatile liquid experiment was a graduated cylinder.

اخترأحد الخيارات

- صح
- خطأ

Usually the rate of reaction increases as temperature decreases

اخترأحد الخيارات

- صح
- خطأ



A suitable equipment that used for measure the temperature is Celicius.



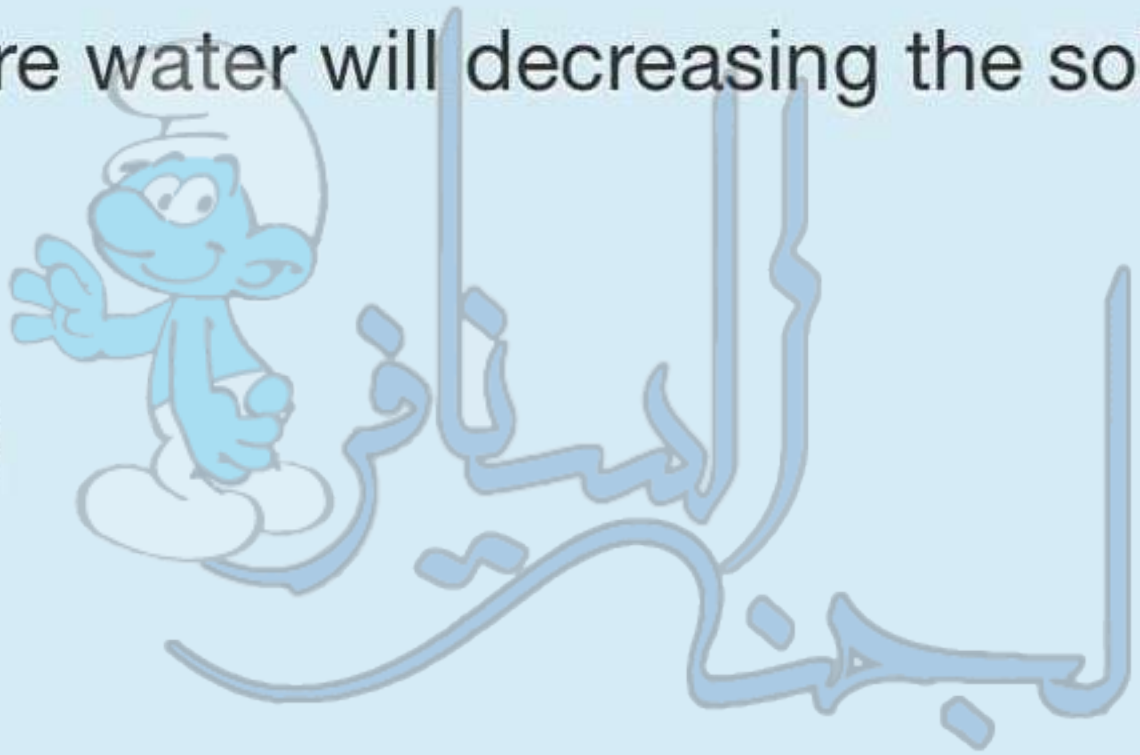
اخترأحد الخيارات

- صح
- خطأ

Adding more water will decreasing the solubility of salts.

Select one:

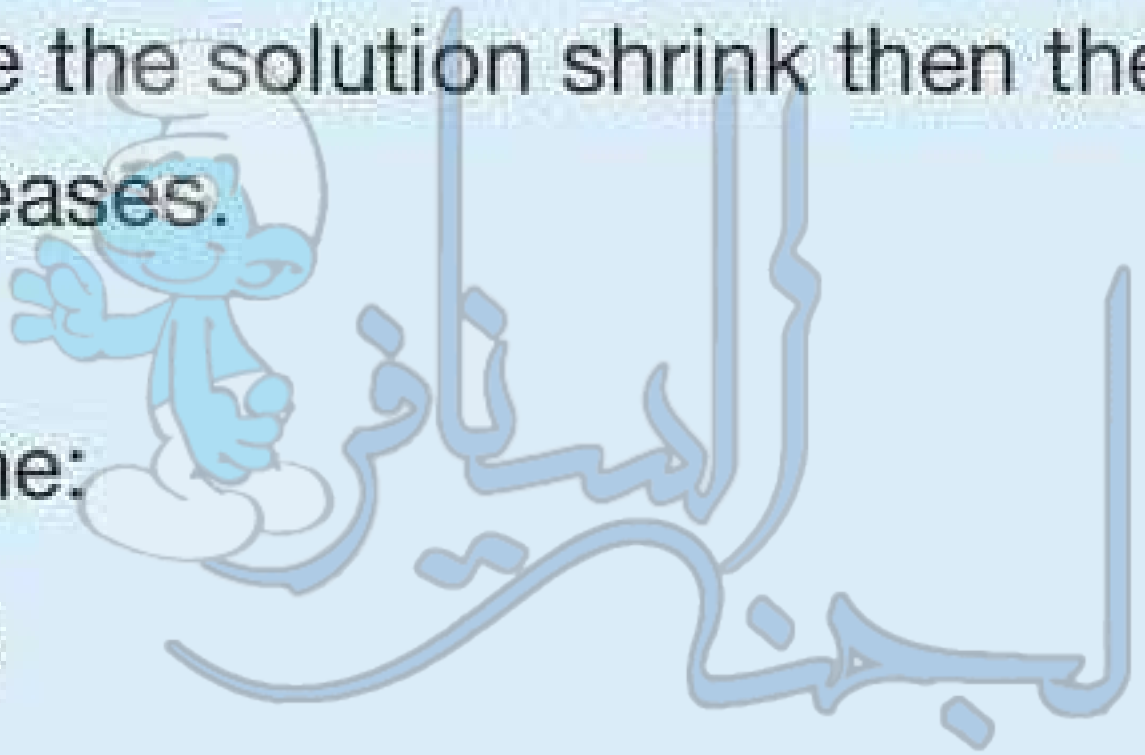
- True
- False



When a part of solid is dissolved in water and this make the solution shrink then the density will decreases.

Select one:

- True
- False



In the formula $MgSO_4 . XH_2O$, the *X* before the H_2O stand for the number of molecules of water in one mole of hydrated salt.

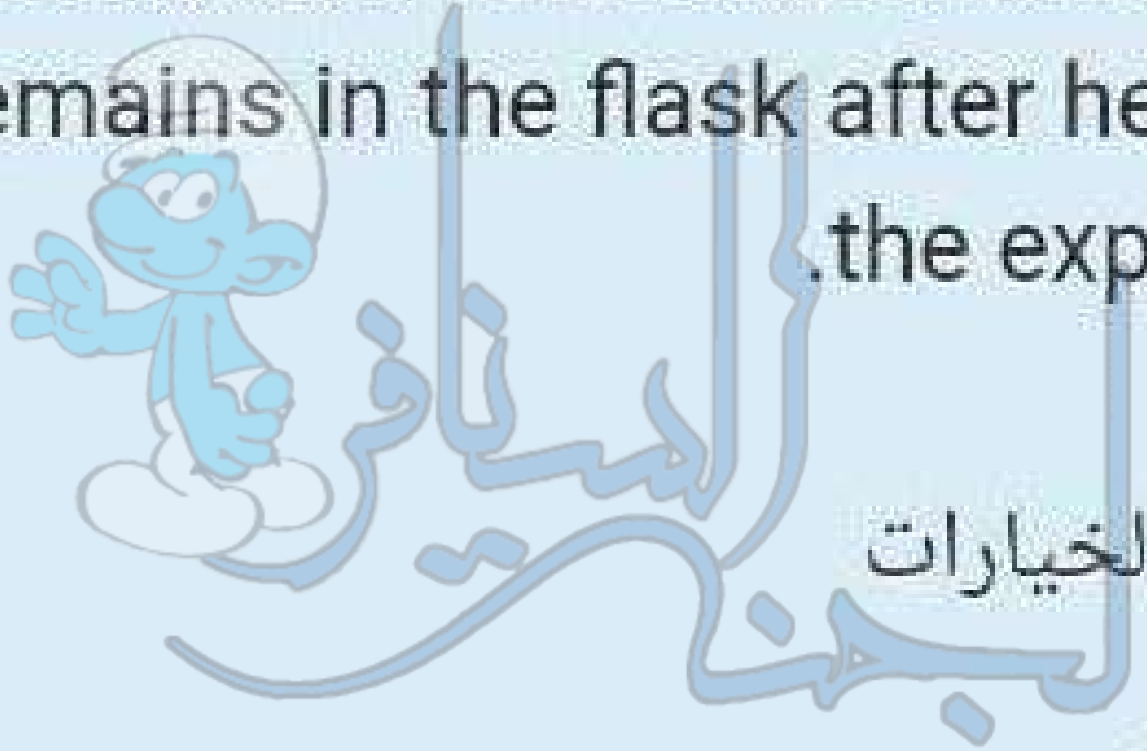
اختر أحد الخيارات

- صح
- خطأ

The variable "V" in the Ideal Gas Law, PV = nRT, represents the volume of volatile liquid that remains in the flask after heating in the experiment.

اخترأحد الخيارات

- صح
- خطأ



It's allowed to withdraw samples form the stock solution bottles with your own pipet.

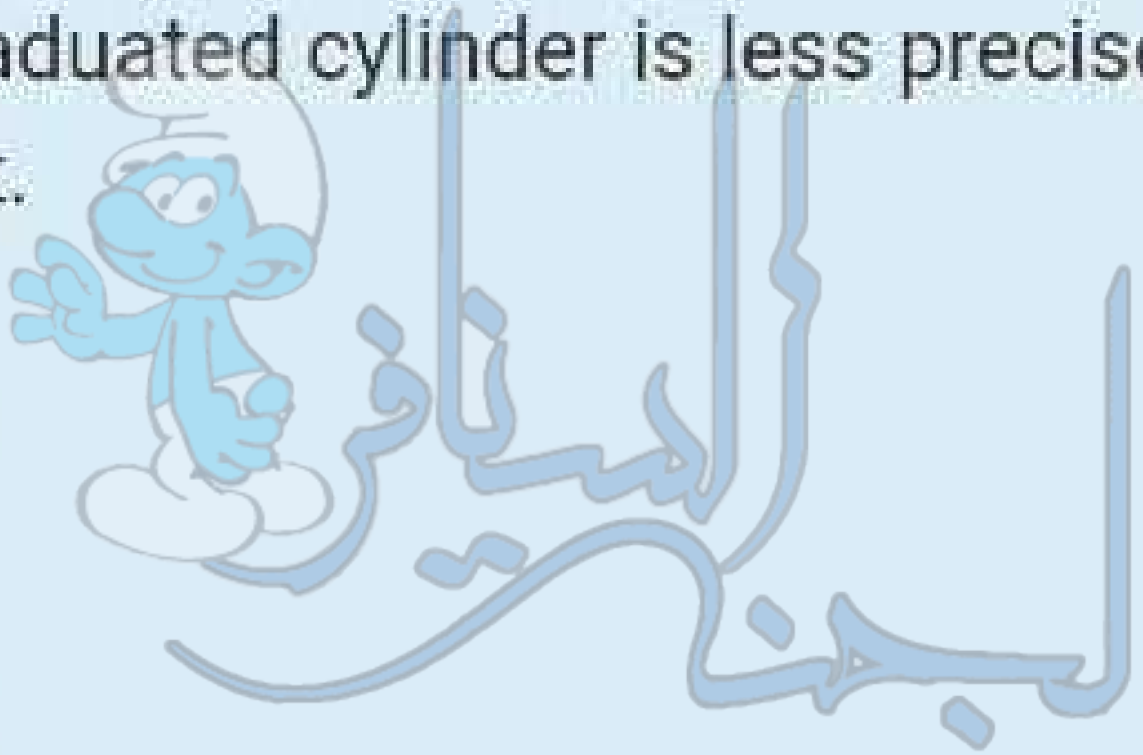
اخترأحد الخيارات

- صح
- خطأ

A 10mL graduated cylinder is less precise than a 10mL pipet.

Select one:

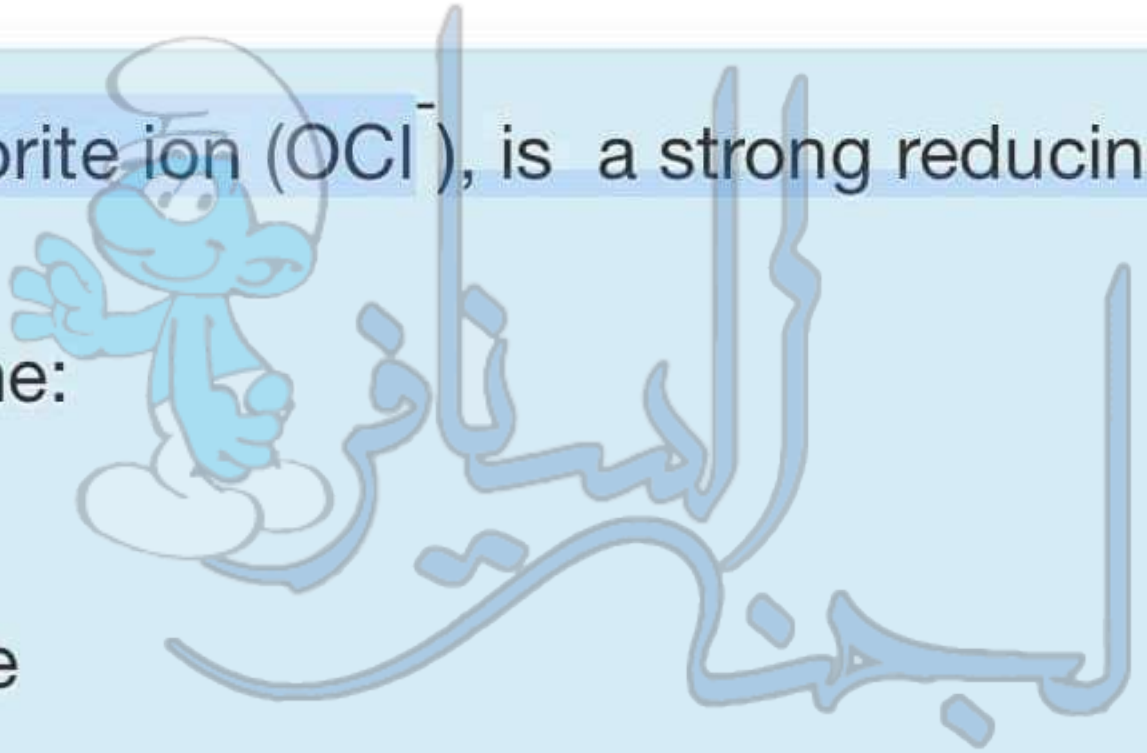
- True
- False



Hypochlorite ion (OCl^-), is a strong reducing agent

Select one:

- True
- False



It is important that the crucible is cooled in desiccator to obtain the best results of the crucible and its contents during water of crystallization experiment .

اخترأحد الخيارات

- صح
- خطأ



All chemical liquids which have boiling point more than 95^{0} c we can use dumas method to find its molecular weight.

اخترأحد الخيارات

- صح
- خطأ

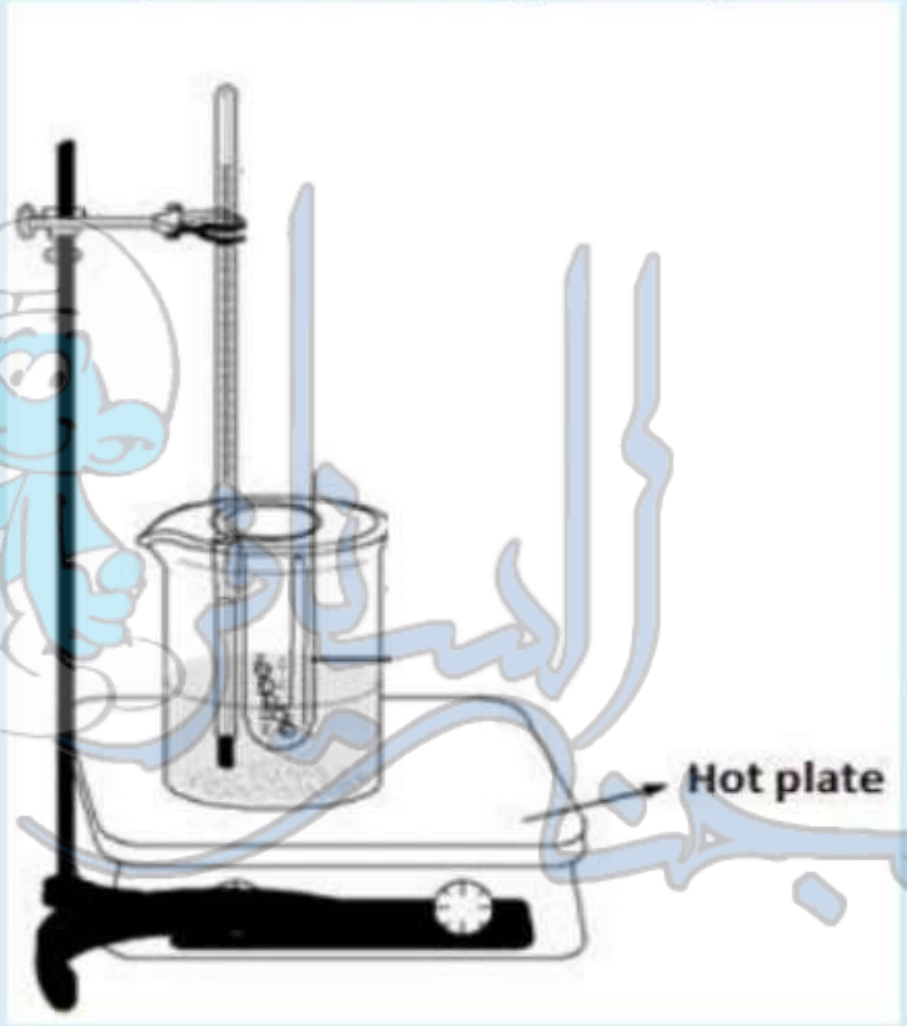


During water of crystallization experiment you indicate the removing of water by expanded , frothed or changing color of the .salt

اخترأحد الخيارات

- صح
- خطأ

This photo shows the correct apparatuses you used in boiling point experiment.



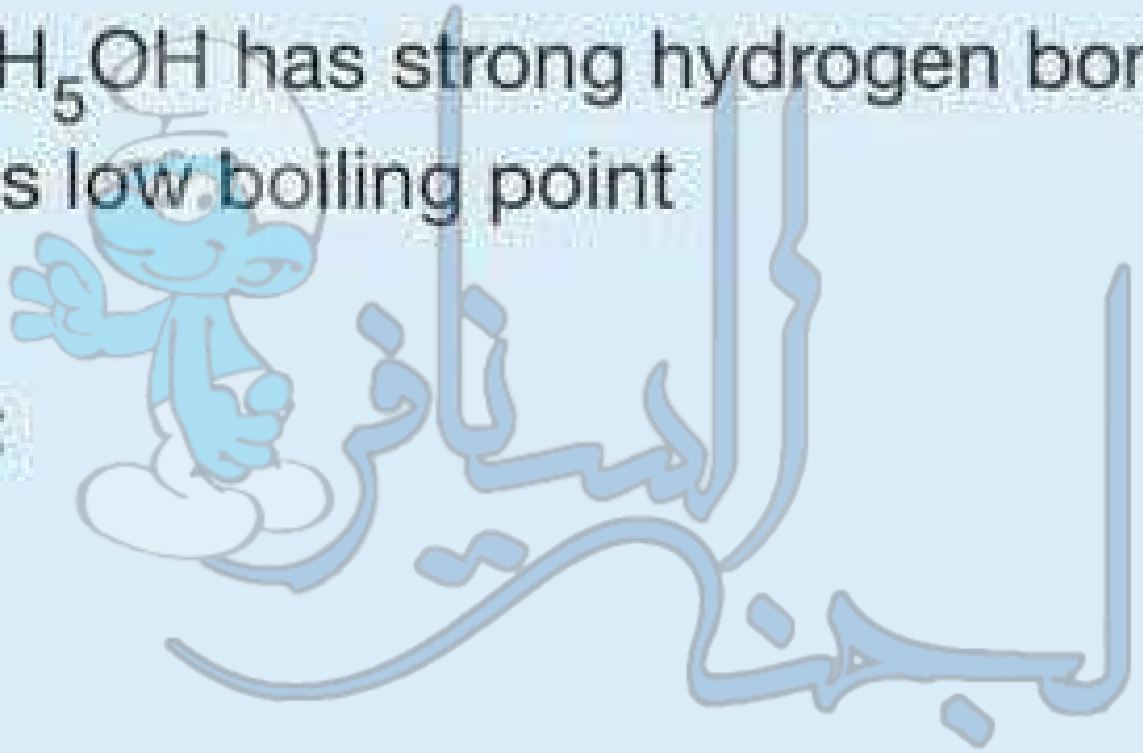
اخترأحد الخيارات

- صح
- خطأ

Ethanol C_2H_5OH has strong hydrogen bonding and hence it has low boiling point

Select one:

- True
- False



During vinegar analysis, a 2.0 mL vinegar sample (d =1.0 g/mL, %(w/w) CH_3COOH= 5.5%, molar mass 60.0 g/mol) was titrated with 0.2M NaOH, If the initial burette reading was 5.0 mL, then the final burette reading (in mL) equals:

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a. 15.17mL

b. 13.17mL

c. 12.17mL

d. 14.17mL

The structure of the precipitate in limiting reagent experiment is:



اخترأحد الخيارات

a. $CaCO_3$

b. NaCl

c. $NaCO_3$

d. $CaCl_2$

The % W/W of NaOCl in a bleach sample equal 8%. The mass of NaOCl in **2L** sample of this bleach (d=1.084 g/mL) equals:

Select one:

- a. 130.1g
- b. 109.3g
- c. 173.4g
- d. 151.76g

For the reaction below, the suitable rate law expression is:

$$2NOCl \rightarrow 2NO + Cl_2$$

اخترأحد الخيارات

a. $R = k[NO]^x[NOCl]^y$

b. $R = k[NOCl]^x$

c. $R = k[NO]^x$

d. $R = k[NO]^x[Cl_2]^y$



The color of the precipitate in limiting reactant experiment is:



Select one:

- a. Blue
- b. Brown
- c. White
- d. Pink

For the reaction below, the suitable rate law expression is:

$$2NOCl \rightarrow 2NO + Cl_2$$

اخترأحد الخيارات

a. $R = k[NO]^x[NOCl]^y$

b. $R = k[NOCl]^x$

c. $R = k[NO]^x$

d. $R = k[NO]^x[Cl_2]^y$



The equipment in the figure is the suitable equipment that you used to heat the hydrated salt

Select one:

- True
- False



Use the following data to calculate the enthalpy change, in ($kJ\ mol^{-1}$), for the reaction . in the box

$$2C_{(s)} + 3H_{2\,(g)} \rightarrow C_2H_{6\,(g)}$$

$C_{(s)} + O_{2\,(g)} \rightarrow CO_{2\,(g)}$ $\Delta H°=$
-393.5 kJ/mol

$1/2O_{2\,(s)} + H_{2\,(g)} \rightarrow H_2O_{(l)}$ $\Delta H°=$
-285.5 kJ/mol

$2C_2H_{6(g)} + 7O_{2\,(g)} \rightarrow 4CO_{2\,(g)} + 6H_2O_{(l)}$ $\Delta H°=$
-3229.6 kJ/mol

اخترأحد الخيارات

a. -27.4

b. 28.7

c. -28.7

d. 393.5

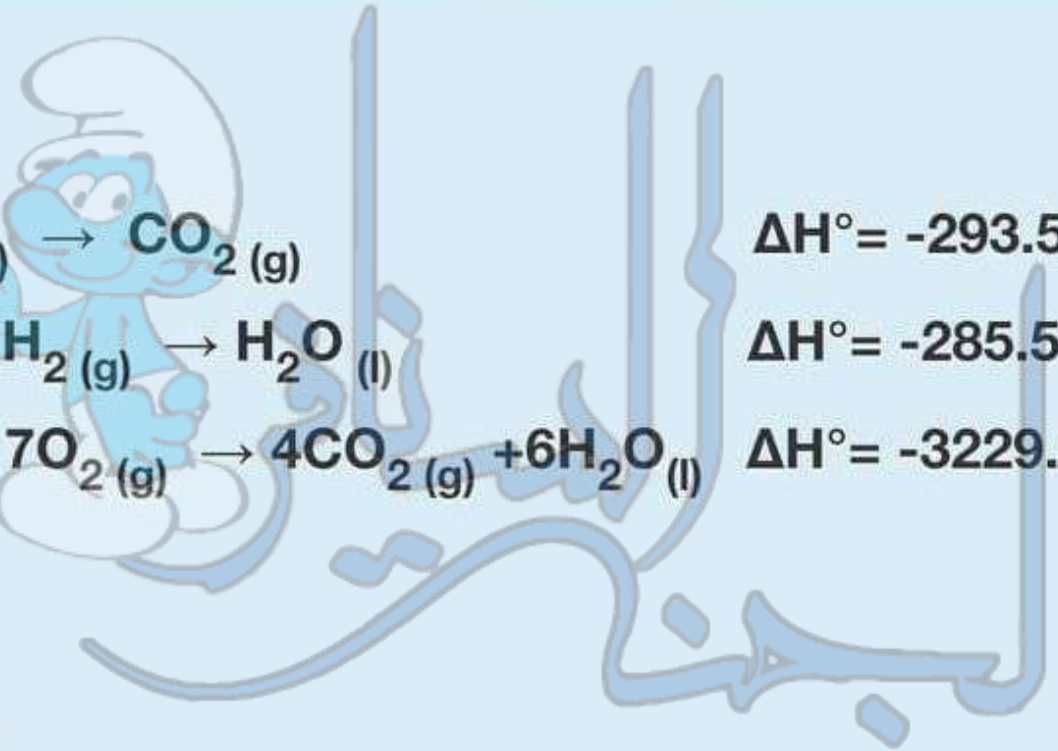
The anhydrous salt was allowed to cool overnight before determining the mass , then the value of (X):

Select one:

- A. No Effect
- B. Increase
- C. Decrease

Use the following data to calculate the enthalpy change, in (kJ mol^{-1}), for the reaction in the box .

$C_2H_{6\,(g)} \rightarrow 2C_{(s)} + 3H_{2\,(g)}$



$C_{(s)} + O_{2\,(g)} \rightarrow CO_{2\,(g)}$ $\Delta H° = -293.5$ kJ/mol

$1/2O_{2\,(s)} + H_{2\,(g)} \rightarrow H_2O_{(l)}$ $\Delta H° = -285.5$ kJ/mol

$2C_2H_{6(g)} + 7O_{2\,(g)} \rightarrow 4CO_{2\,(g)} + 6H_2O_{(l)}$ $\Delta H° = -3229.6$ kJ/mol

Select one:

- a. -171.3
- b. -28.7
- c. 28.7
- d. 171.3

During experiment of chemical kinetics for the reaction $H_2O_2 + 2I^- + 2H^+ \rightarrow I_2 + 2H_2O$, :the rate law expression is

$Rate = k[H_2O_2]^x [I^-]^y$. Use the data in table to calculate y

Rate (M/s)	M $[H_2O_2]$	M $[I^-]$	#Run
0.24	0.064	0.22	1
0.96	0.064	0.88	2
0.24	0.032	0.22	3

اختر أحد الخيارات

a. y=1

b. y=2

c. y=3

d. y=0

During experiment of chemical kinetics for the reaction $H_2O_2 + 2I^- + 2H^+ \rightarrow I_2 + 2H_2O$, the rate law expression is:

Rate = $k [H_2O_2]^Y [I^-]^X$. Use the data in table to calculate y

Run#	$[I^-]$ M	$[H_2O_2]$ M	Rate (M/s)
1	0.22	0.064	0.24
2	0.88	0.064	0.96
3	0.22	0.032	0.24

Select one:

- a. y=1
- b. y=3
- c. y=2
- d. y=0

Q5- The diagram beside shows data concerning experiment of determination of enthalpy of neutralization of the reaction:

$NaOH + HCl$ ----------$H_2O + NaCl$

$\Delta H = -71.5 KJ/mol$ .What is the maximum temperature reached during reaction, if (0.06 moles) of NaOH used in exp. (specific heat= 4.12 J/g.◦c):

Select one:

- 1. 41.0 ◦c
- 2. 35.9◦c
- 3. 20.02 ◦c
- 4. 45.2 ◦c

Consider the data shown in the table below. Which of those two metals will absorb less heat when temperature changes to 100 °C?

Metal	A	B
Mass (g)	100.00	100.00
S (J/g °C)	0.258	0.885

اختراحد الخيارات

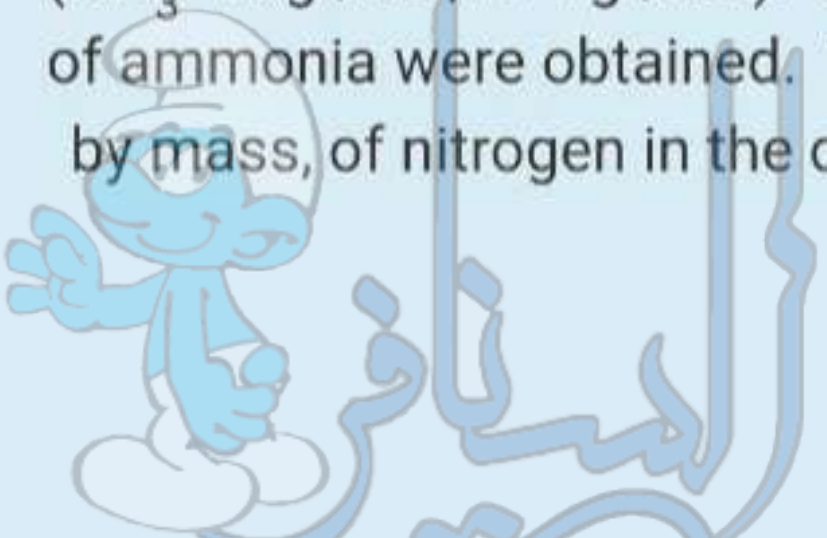
a. B

b. none of them

c. Both

d. A

A 4.294 g sample of a compound was analyzed for nitrogen N(N=14g\mol). In the procedure, all the nitrogen present was completely converted to ammonia (NH_3=17g\mol, H=1g\mol). 3.196 grams of ammonia were obtained. The percent, by mass, of nitrogen in the compound is :therefore



اختر أحد الخيارات

a. 49.7%

b. 61.2%

c. 28.3%

% d. 36.0

What might happen if you mixed a strong acid with an equally strong base?

اخترأحد الخيارات

A. The base would destroy the acid.

B. The acid would destroy the base.

C. You'd wind up with a pH-neutral solution.

D. You would see an explosive chemical reaction.

A 5.294 g sample of a compound was analyzed for nitrogen N (N=17g\mol). In the procedure, all the nitrogen present was completely converted to ammonia (NH_3=17g\mol,H=1g\mol) 3.196 grams of ammonia were obtained. The percent, by mass, of nitrogen in the compound is therefore:

Select one:

- a. 28.3%
- b. 36.0 %
- c. 61.2%
- d. 49.7%

Using the blue flame in hydrated salt experiment , then the value of (H_2O %) will be:

Select one:

- A. Decrease
- B. No effect
- C. Increase

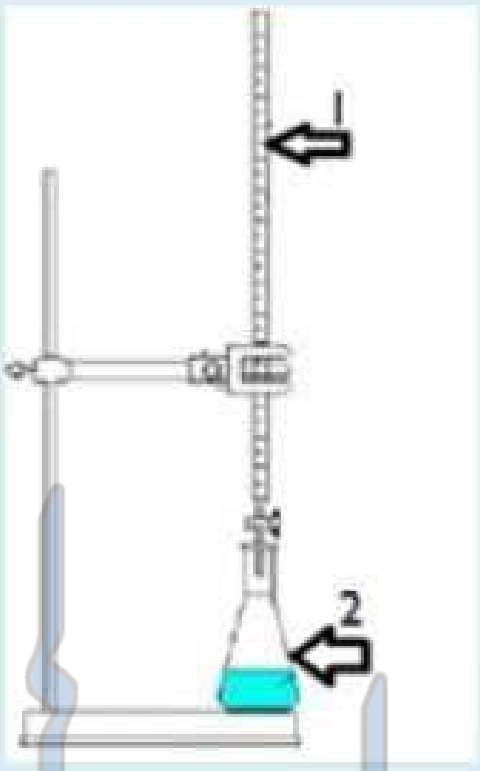
What is the name of the following glassware?



Select one:

- A. burette
- B. Desiccator
- C. Volumetric Flask
- D. Funnel

Consider the set up on the right which used during vinegar analysis. The names of solution filled in equipment 1 and equipment 2 are:



Select one:

- a. in 1 HCl solution and in 2 NaOH solution
- b. in 1 CH_3COOH (vinegar solution) and in 2 HCl solution
- c. in 1 NaOH solution and in 2 CH_3COOH (vinegar solution)
- d. in 1 CH_3COOH (vinegar solution) and in 2 NaOH solution

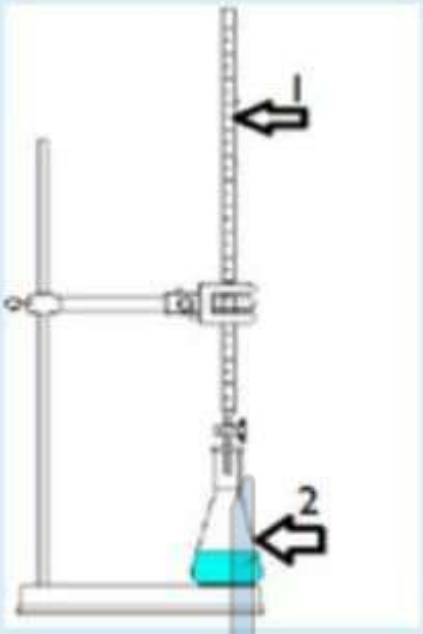
A 9.294 g sample of a compound was analyzed for nitrogen N (N=14g\mol). In the procedure, all the nitrogen present was completely converted to ammonia (NH_3=17g\mol ,H=1g\mol). 3.196 grams of ammonia were obtained. The percent, by mass, of nitrogen in the compound is :therefore



اخترأحد الخيارات

- % a. 36.0
- b. 49.7%
- c. 28.3%
- d. 61.2%

Consider the set up on the right which used during vinegar analysis. The names of equipment 2 and 1 are



Select one:

- a. 1-flask 2-burette
- b. 1-beaker
  2-test tube
- c.
  2-Flask 1-Burette
- d.
  1-flask
  2-test tube

During limiting reactant experiment, when 30.0 mL of 0.55 M Na_2CO_3 (molar mass 105.99 g/mol) solution were mixed with 15.0 mL of 1.2 M $CaCl_2$ (111.0 g/mol) solution then 0.865 g of $CaCO_3$ (100.09) were obtained. The percentage yield of this reaction equal:

$$Na_2CO_3 + CaCl_2 \rightarrow CaCO_3 + 2NaCl$$

Select one:

- a. 40.3%
- b. 46.3%
- c. 58.5%
- d. 52.4%

A20.0 mL bleach sample was diluted to 100 mL in a volumetric flask. A 25.0 mL of this solution required 12.5 mL of 0.30 M $Na_2S_2O_3$ to reach the end point of titration. Calculate the mass percent of NaClO in the original sample (Molar mass NaClO = 74.5 g/mol). Assume the density of bleach solution is 1.084g/mL.

Select one:

- a. 10.30%
- b. 2.58%
- c. 1.29%
- d. 5.15%

Use the following data to calculate the enthalpy change, in (kJ mol^{-1}), for the reaction in the box .

$$2C_{(s)} + 3H_{2(g)} \rightarrow C_2H_{6(g)}$$

$$C_{(s)} + O_{2(g)} \rightarrow CO_{2(g)}$$
$$\Delta H^\circ = -393.5 \text{ kJ/mol}$$

$$1/2O_{2(s)} + H_{2(g)} \rightarrow H_2O_{(l)}$$
$$\Delta H^\circ = -285.5 \text{ kJ/mol}$$

$$2C_2H_{6(g)} + 7O_{2(g)} \rightarrow 4CO_{2(g)} + 6H_2O_{(l)}$$
$$\Delta H^\circ = -3229.6 \text{ kJ/mol}$$

Select one:

- a. -27.4
- b. -28.7
- c. 28.7
- d. 393.5

A40.0 mL bleach sample was diluted to 100 mL in a volumetric flask. A 25.0 mL of this solution required 12.5 mL of 0.30 M $Na_2S_2O_3$ to reach the end point of titration. Calculate the mass percent of NaClO in the original sample (Molar mass NaClO = 74.5 g/mol). Assume the density of bleach solution is 1.084g/mL.

Select one:

- a. 1.29%
- b. 5.15%
- c. 10.30%
- d. 2.58%

A compound has the empirical formula of **SF_4 (108g\mol)**. At 20 °C ,0.200 g of the gaseous compound occupies a volume of 22.1 ml with 1.02 atm. What is the molecular formula of this compound ?(R=0.08206L.atm\K.mol)



اختر أحد الخيارات

a. S_4F_{16}

b. S_2F_8

c. S_3F_{12}

d. SF_4

Use the following data to calculate the enthalpy change, in ($kJ\ mol^{-1}$), for the reaction in the box.

$$C_2H_{6\,(g)} \rightarrow 2C_{(s)} + 3H_{2\,(g)}$$

$$C_{(s)} + O_{2\,(g)} \rightarrow CO_{2\,(g)}$$
$$\Delta H° = -293.5\ kJ/mol$$

$$1/2O_{2\,(s)} + H_{2\,(g)} \rightarrow H_2O_{(l)}$$
$$\Delta H° = -285.5\ kJ/mol$$

$$2C_2H_{6(g)} + 7O_{2\,(g)} \rightarrow 4CO_{2\,(g)} + 6H_2O_{(l)}$$
$$\Delta H° = -3229.6\ kJ/mol$$

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a. 171.3

b. -171.3

c. 28.7

d. -28.7

A5.0 mL bleach sample was diluted to 100 mL in a volumetric flask. A 25.0 mL of this solution required 12.5 mL of 0.30 M $Na_2S_2O_3$ to reach the end point of titration. Calculate the mass percent of NaClO in the original sample (Molar mass NaClO = 74.5 g/mol). Assume the density of bleach solution is 1.084g/mL.

Select one:

- a. 1.29%
- b. 2.58%
- c. 10.30%
- d. 5.15%

For the reaction below, the suitable rate law expression is:

$$2NO + Cl_2 \rightarrow 2NOCl$$

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a. $R = k[NO]^x[Cl_2]^y$

b. $R = k[NO]^x$

c. $R = k[NO]^x[NOCl]^y$

d. $R = k[NOCl]^x$

The name of technique that was used to separate the precipitate from liquid in limiting reagent experiment is:



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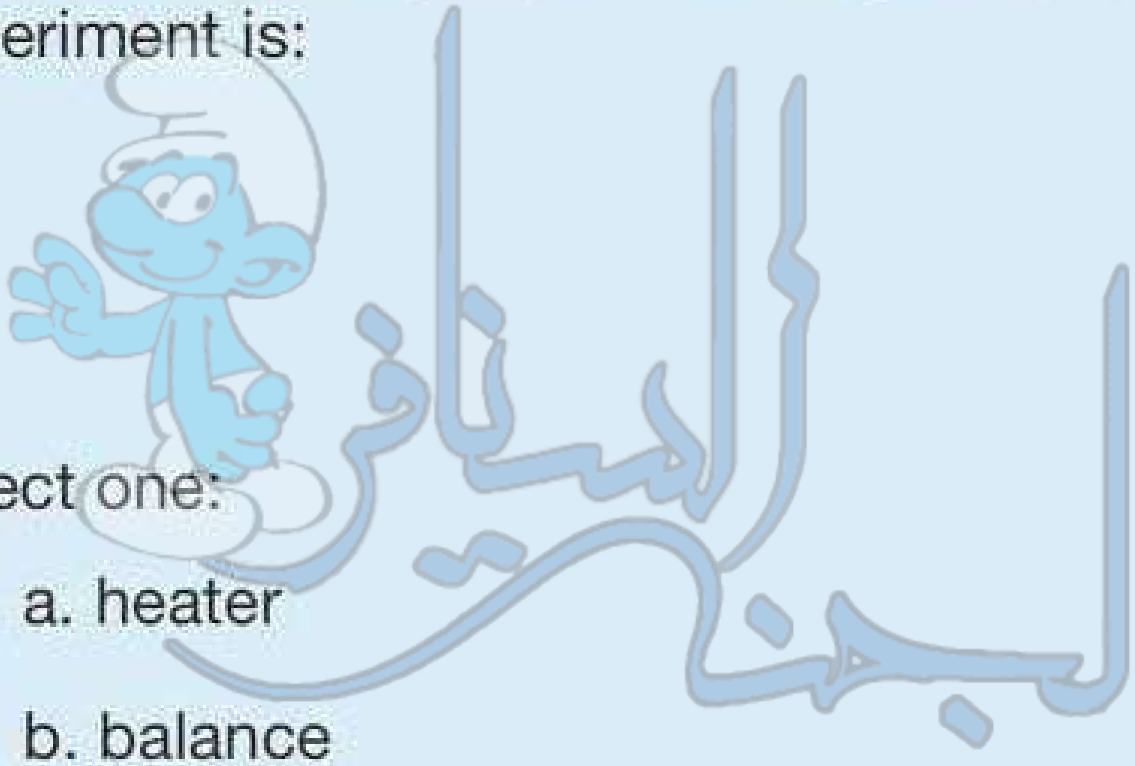
a. Boiling solution

b. Suction filtration

c. Precipitation

d. evaporator water

The name of equipment that was used to dry the wet precipitate in limiting reagent experiment is:



Select one:

- a. heater
- b. balance
- c. thermometer
- d. oven